

LAINIO, Sweden

WMO: 020860

Lat: 67.761N

Lon: 22.352E

Elev: 1040

StdP: 14.15

Time Zone: 1.00 (EUC)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.8	-19.5	-28.2	1.2	-23.7	-23.5	1.6	-19.3	24.6	28.0	21.6	27.3	2.4	320	0.774

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	71.9	58.1	68.0	56.1	64.5	54.5	59.7	67.9	57.8	65.1	55.9	62.3	7.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.5	70.7	62.1	54.7	66.1	60.0	52.7	61.4	58.4	26.7	67.5	25.4	65.4	24.2	62.1	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	17.5	15.1	DB	-29.4	77.1	7.4	2.6	-34.7	78.9	-39.0	80.4	-43.2	81.9	-48.6	83.7	(4)
(5)			WB	-29.5	62.3	7.4	2.8	-34.9	64.3	-39.2	66.0	-43.4	67.6	-48.7	69.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.1	7.7	11.8	18.8	27.5	39.4	51.3	54.7	50.6	41.2	29.3	16.7	10.9	(6)	
(7)		DBStd	19.05	14.88	12.86	8.70	8.24	6.39	7.03	5.31	6.01	5.85	9.73	11.18	14.76	(7)	
(8)		HDD50	7629	1310	1068	968	675	337	66	15	67	270	642	1000	1211	(8)	
(9)		HDD65	12745	1775	1488	1433	1125	794	413	324	447	713	1107	1450	1676	(9)	
(10)		CDD50	364	0	0	0	0	9	104	160	85	6	0	0	0	(10)	
(11)		CDD65	6	0	0	0	0	0	2	3	1	0	0	0	0	(11)	
(12)		CDH74	39	0	0	0	0	0	11	25	3	0	0	0	0	(12)	
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14)	Wind	WSAvg	6.2	5.4	5.3	6.2	6.7	7.2	7.7	7.2	6.2	5.8	6.4	5.1	5.4	(14)	
(15)	Precipitation	PrecAvg	21.7	1.3	1.1	1.0	1.1	1.7	2.6	3.4	2.7	2.0	1.9	1.7	1.4	(15)	
(16)		PrecMax	28.5	2.4	2.1	3.0	3.0	3.1	4.5	6.2	4.4	4.9	3.6	3.3	3.7	(16)	
(17)		PrecMin	15.4	0.2	0.2	0.2	0.4	0.4	0.5	1.2	0.8	0.4	0.8	0.5	0.2	(17)	
(18)		PrecStd	3.5	0.5	0.6	0.6	0.7	0.8	1.1	1.4	1.0	1.2	0.8	0.7	0.7	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	37.5	40.2	48.6	68.3	76.1	77.7	74.4	60.6	49.9	37.2	37.4	(19)	
(20)			MCWB	33.5	33.5	34.5	39.8	52.8	59.4	63.6	60.3	52.5	43.2	34.2	33.9	(20)	
(21)		2%	DB	34.5	33.7	36.0	44.2	60.4	71.7	71.6	67.9	57.0	46.5	34.7	34.3	(21)	
(22)			MCWB	31.2	31.2	31.5	37.1	48.2	57.9	58.3	57.3	50.5	42.5	32.8	31.7	(22)	
(23)		5%	DB	31.4	31.3	33.3	41.4	54.6	67.5	68.0	63.7	53.5	44.0	33.0	32.0	(23)	
(24)			MCWB	29.0	29.7	30.6	35.7	44.7	54.8	56.6	55.2	48.6	40.6	31.6	30.4	(24)	
(25)		10%	DB	28.2	28.8	30.9	38.7	50.0	63.3	64.5	60.5	50.2	41.3	31.7	29.6	(25)	
(26)			MCWB	26.6	27.5	28.8	33.9	42.0	52.2	55.1	53.6	46.0	38.7	30.7	28.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.9	34.2	34.7	41.1	53.5	61.9	65.1	61.4	55.5	45.7	35.2	34.5	(27)	
(28)			MCDWB	36.2	36.3	38.7	45.2	68.0	71.0	73.6	71.2	57.7	47.0	36.0	36.6	(28)	
(29)		2%	WB	31.3	31.7	32.3	38.1	49.2	59.0	60.0	58.9	52.3	43.3	33.2	32.3	(29)	
(30)			MCDWB	33.9	33.1	34.8	42.9	59.4	68.7	67.5	65.0	55.8	45.4	34.2	33.8	(30)	
(31)		5%	WB	29.3	29.8	30.8	36.2	45.3	56.1	58.0	57.0	49.0	40.6	32.0	30.5	(31)	
(32)			MCDWB	31.5	31.3	33.0	40.2	53.4	66.2	65.7	62.0	52.3	43.5	32.8	31.8	(32)	
(33)		10%	WB	26.5	27.5	29.0	34.7	42.7	53.4	56.2	54.8	46.7	38.9	30.5	28.2	(33)	
(34)			MCDWB	28.1	28.8	30.9	38.1	49.1	61.2	62.7	59.0	49.4	41.6	31.7	29.7	(34)	
(35)	Mean Daily Temperature Range			MDBR	13.5	13.8	14.2	14.3	13.2	13.6	13.4	12.4	11.6	10.3	11.6	14.0	(35)
(36)		5% DB	MCDBR	11.7	12.3	15.1	15.9	21.8	20.4	18.6	17.0	15.8	12.1	8.8	14.0	(36)	
(37)			MCWBR	10.6	10.8	12.2	10.9	12.4	10.8	8.9	8.9	9.5	9.2	8.1	12.9	(37)	
(38)		5% WB	MCDWB	11.6	11.5	12.5	13.2	21.1	18.8	15.9	13.7	12.9	10.6	8.2	13.5	(38)	
(39)			MCWBR	10.6	10.3	10.5	9.4	12.4	11.0	8.5	8.0	8.2	8.7	7.7	12.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.180	0.242	0.278	0.308	0.327	0.333	0.339	0.324	0.294	0.263	0.346	N/A	(40)	
(41)		taud		1.927	2.114	2.210	2.240	2.445	2.492	2.487	2.511	2.569	2.353	2.796	N/A	(41)	
(42)		Ebn at Noon		43	182	243	267	276	277	271	263	246	179	58	0	(42)	
(43)		Edh at Noon		3	15	24	32	29	29	28	24	18	12	4	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg		14	167	648	1262	1544	1594	1493	1082	583	196	32	0	(44)	
(45)		RadStd		1	18	37	56	148	110	146	80	72	31	4	0	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page